CS 341

LAB ASSIGNMENT

3,4 stage pipelined processor Implementation

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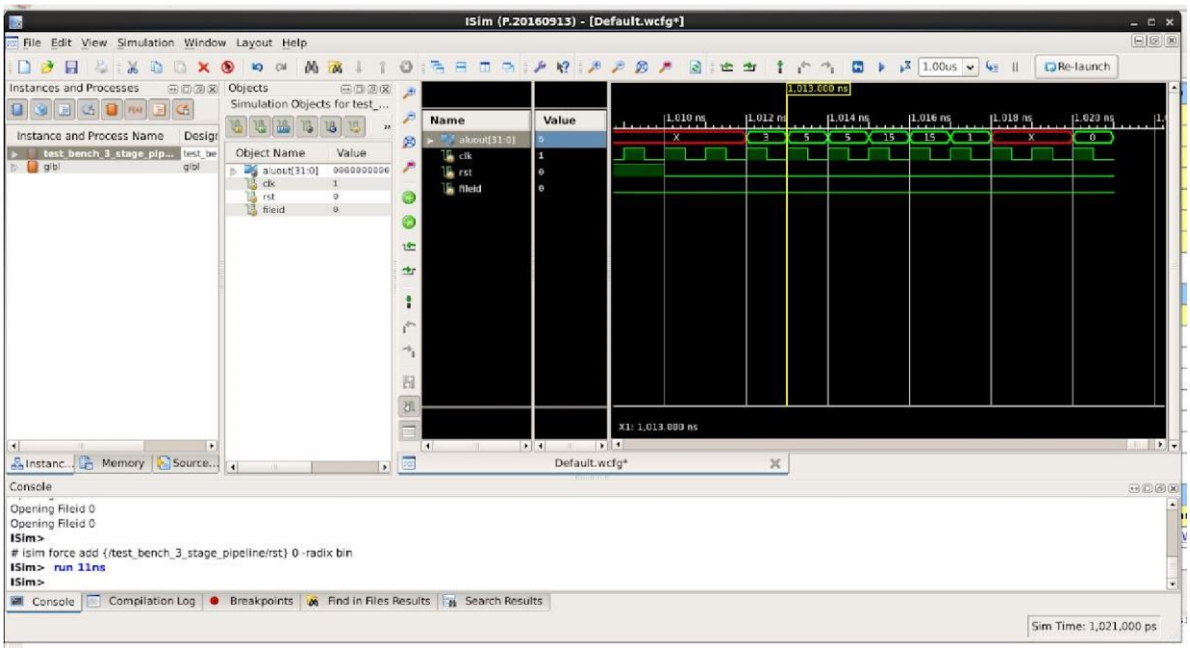
3-stage pipelined processor implementation

- Input to the main module contains clk, rst and fileid. Instruction memory is read from the file imem_test0.txt.
- Output from ALU module is the output for the main module.
- There are several modules like PC_Increment, MUXes, ALU, memory, regfile and 2 modules for pipeline registers.
- Below is the simulation of testbench_3_stage_pipeline.v.



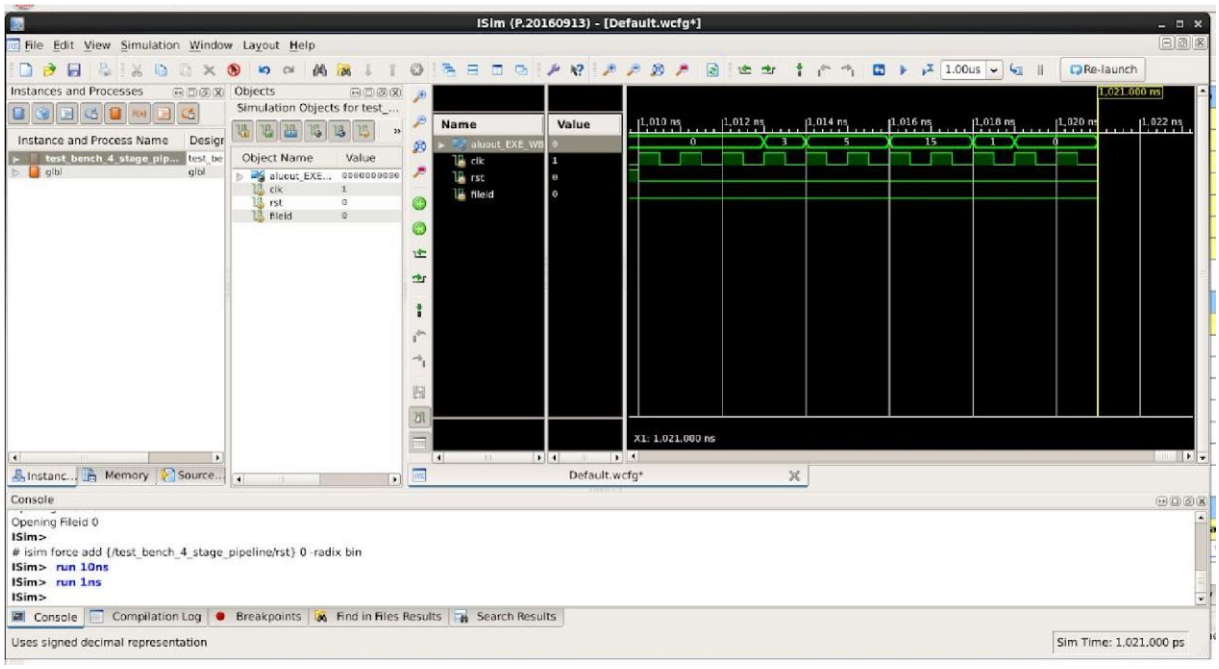
- No. of LUT Slices (Area) = 2694

- Max operating Frequency = 6.901 MHz

- Below is the Screenshot for Area and Max Frequency.

4-stage pipelined Processor Implementation

- Input to the main module contains clk, rst and fileid. Instruction memory is read from the file imem_test0.txt.
- Output from EXE_WB module is the output for the main module.
- There are several modules like PC_Increment, MUXes, ALU, memory, regfile and 3 modules for pipeline registers.
- Below is the simulation of testbench_4_stage_pipeline.v



- No. of LUT Slices (Area) = 2751

- Max operating Frequency = 68.930 MHz